Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
4 yards = _____ inches

A. 12

B. 48

C. 144

D. 432

2. Which numbers are equivalent to 2.75? Circle all that apply.

A. $2\frac{3}{4}$

B. $\frac{11}{4}$

C. $2\frac{7}{10}$

D. $\frac{4}{11}$

E. $2\frac{75}{100}$

F. $\frac{275}{10}$

3. Which list shows the ratios in order from least to greatest?

A. $\frac{4}{5}$, 3 : 4, 7 to 10

B. 3 : 4, 7 to 10, $\frac{4}{5}$

C. 7 to 10, 3 : 4, $\frac{4}{5}$

D. 7 to 10, $\frac{4}{5}$, 3 : 4

4. A hockey team has a win to loss ratio of 3 : 4. Which could be the team's record?

A. 6 wins and 14 games played

B. 8 wins and 6 losses

C. 9 wins and 12 games played

D. 12 wins and 15 losses

5. A store sells packs of granola bars. Which pack is the best buy?

A. 6 bars for $4.80

B. 10 bars for $7.50

C. 15 bars for $11.25

D. 12 bars for $8.40

6. Which rate is **not** equal to 1?

A. $\frac{12 \text{ inches}}{1 \text{ foot}}$

B. $\frac{1 \text{ gallon}}{4 \text{ quarts}}$

C. $\frac{1 \text{ pound}}{12 \text{ ounces}}$

D. $\frac{1{,}000 \text{ meters}}{1 \text{ kilometer}}$

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to a conveyor belt moving 18 feet per minute.

7a. 6 yards per minute	**YES**	**NO**
7b. 216 inches per minute	**YES**	**NO**
7c. 1,080 feet per second	**YES**	**NO**
7d. 1,080 feet per hour	**YES**	**NO**

8. Simplify the complex fraction $\frac{6}{\frac{3}{4}}$.

A. $4\frac{1}{2}$

B. 8

C. $\frac{1}{8}$

D. $\frac{3}{24}$

9. Tess rode her bike $2\frac{1}{2}$ miles in 15 minutes. Assume she rode the entire time at the same rate. What was her speed in miles per hour?

A. 10 miles per hour

B. 6 miles per hour

C. $\frac{1}{6}$ mile per hour

D. $37\frac{1}{2}$ miles per hour

Form D

Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

Name _____ Period _____ Date _____

Part II – Constructed Response

Complete each conversion.

1. 5 gallons = _____ quarts

2. 3 kilometers = _____ centimeters

3. 36 fluid ounces = _____ cups

4. 420 seconds = _____ minutes

Convert each mixed number to a decimal.

5. $2\frac{5}{8}$

6. $3\frac{2}{9}$

Convert each decimal to a fraction or mixed number in simplest form.

7. 0.85

8. 3.125

9. Simplify the ratio 30 : 42. Write the simplified ratio in all three forms.

10. Fourteen of 21 tomato plants in a community garden have ripe tomatoes. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers): Explanation:	Ratio (numbers): Explanation:	Ratio (numbers): Explanation:

Find the unit rate.

11. Liam read 30 pages in 45 minutes. Write his unit rate in pages per minute. Then write it in minutes per page.

12. $\dfrac{3\frac{1}{3} \text{ miles}}{\frac{5}{6} \text{ hour}}$

13. A 24-ounce jar of peanut butter costs $4.32. A 40-ounce jar costs $6.80. Which jar is the better buy? Show all work necessary to justify your answer.

14. A jogger runs at a rate of 6 miles per hour. Write this rate as feet per second. Show all work necessary to justify your answer.

15. Ava swam 1.2 kilometers in 30 minutes. Ben swims at a rate of 45 meters per minute. Who swims faster? Show all work necessary to justify your answer.

16. Owen walked $2\frac{1}{4}$ miles in $\frac{3}{4}$ hour. Assume he walked the entire time at the same rate. What was his speed in miles per hour? Priya answered this question, but made a mistake. Explain Priya's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{2\frac{1}{4} \text{ miles}}{\frac{3}{4} \text{ hour}}$

$\frac{2\frac{1}{4} \text{ miles}}{\frac{3}{4} \text{ hour}} = \frac{3}{4} \div \frac{3}{4}$

$= \frac{3}{4} \cdot \frac{4}{3}$

$= \frac{12}{12}$

= 1 mile per hour

Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

Name _____ Period _____ Date _____

Part III – Problem Solving

The garden club has 6 parents and 15 students. They pull weeds at a park to raise money for new seeds, and a sponsor pays for each bag of weeds they fill. Each parent earned $4.00 per bag and filled one bag every 12 minutes. Each student earned $3.00 per bag and filled 3 bags every 45 minutes. What is the ratio of the amount of money the parents raised in one hour to the amount of money the students raised in one hour?